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Your Brain's Alertness or Sleep
Patterns?

Brandon Wang

How to Connect a
Single-Pole Switch to a
Light Bulb

Denise Lee

Why Do We Dream?
Darren Wang

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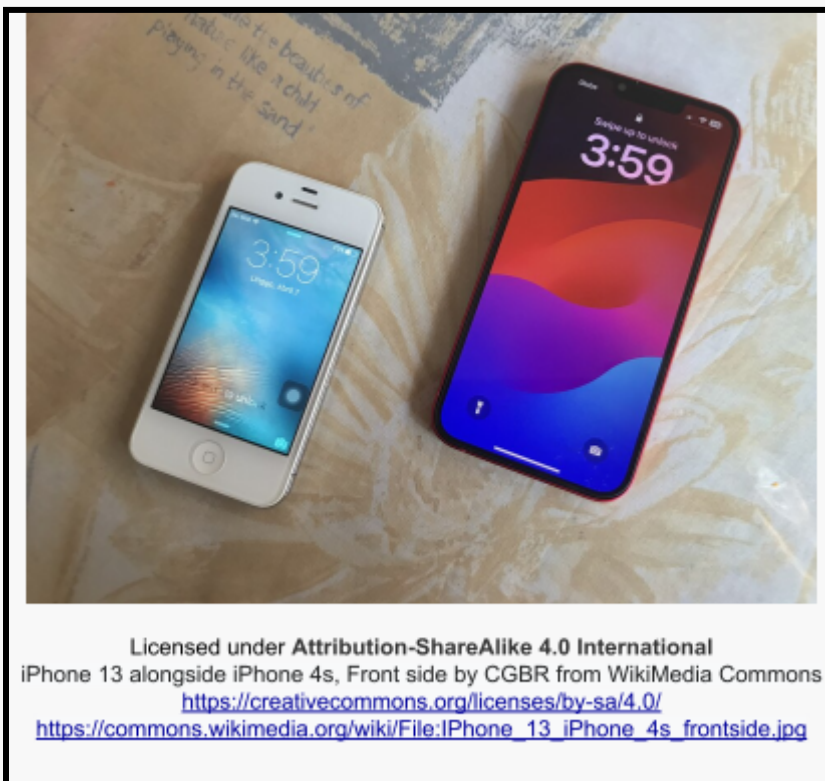
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How Does Blue Light Affect Your Brain's Alertness or Sleep Patterns?

Brandon Wang

Blue light is emitted from natural sunlight and artificial sources like phones and computer screens. It can play a significant role in regulating human circadian rhythms, which is the internal clock of the body, controlling sleep and wake cycles. Exposure to blue light during daylight hours stimulates specific retinal cells, which then communicate with the brain's suprachiasmatic nucleus, a central component in managing circadian rhythms. This process promotes alertness and can improve cognitive performance during daylight hours.



However, exposure to blue light late in the evening or at night can disrupt normal sleep patterns. Research indicates that nighttime blue light exposure can delay sleep onset, reduce overall sleep duration, and negatively impact sleep quality. This is extremely relevant in today's society, where screens are used during the evening. Despite these negative effects, blue light is also used

for light therapy, which treats circadian rhythm disorders and seasonal affective disorder. By mimicking natural sunlight, controlled blue light exposure can help realign disrupted sleep patterns and enhance mood.

In conclusion, blue light has a huge influence on the brain's alertness and sleep regulation. While daytime exposure supports wakefulness, excessive blue light at night can affect healthy sleep, emphasizing the importance of mindful screen use, particularly in the hours before bedtime.

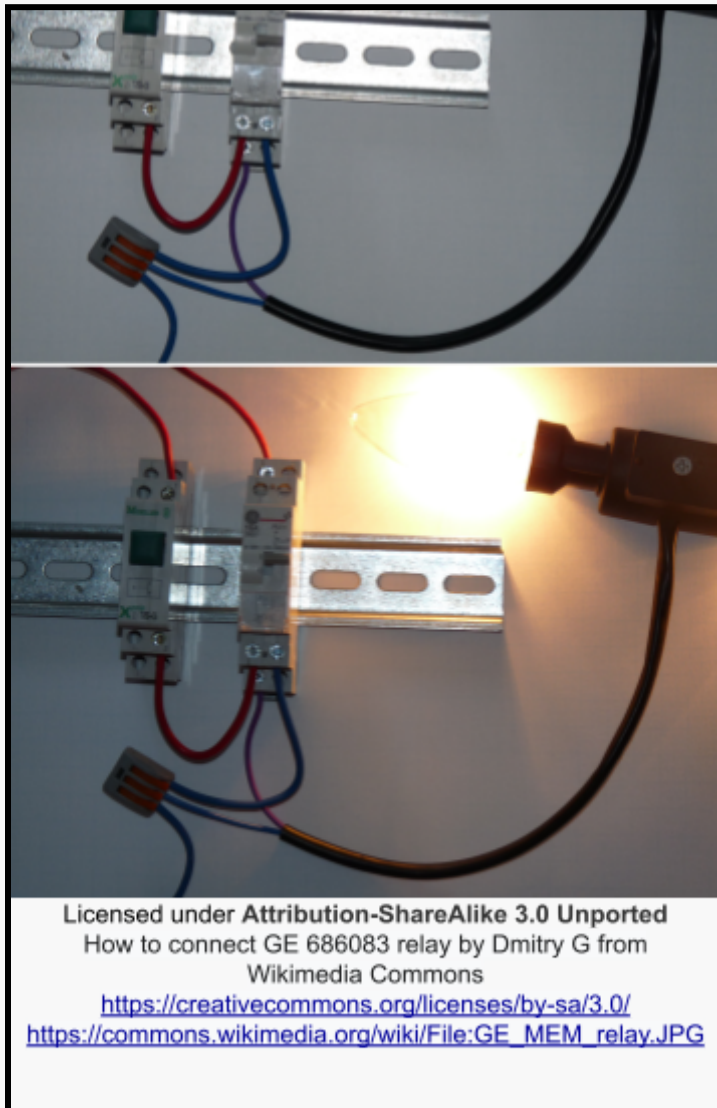
How to Connect a Single-Pole Switch to a Light Bulb?

Denise Lee

What's the one thing we all depend on, every single day? Sure, we need food, water, and air to survive, but just like plants that need sunlight to grow. We need lights to work, cook, clean, study, and conduct everyday tasks. Imagine your light suddenly goes out when you're at home, and you find out it's a wiring issue. The first thing you may do is call an electrician. If you knew how to rewire your switch to your light, you could save money, time, and the frustration that comes with waiting for an electrician to save the day. First, I'll go over the tools and materials you'll need. Then, I'll explain basic electrical concepts, including how electricity flows through the circuit. Finally, I'll teach you how to connect the wires from the switch to the light.

First, let's assemble the tools you'll need to wire the switch and the light. You'll need to get gloves to prevent yourself from getting poked by the wires and in case of any shock. You'll need to get a Compact Wire Stripper/Cutter and a flat-head screwdriver. Next, you want to buy some wire caps: The yellow caps are used for smaller wire connections with 1-3 wires. The red caps are used to connect more wires together up to 5 wires. Lastly, it's helpful to have extra wires in case any of your wires are damaged or are too short to reuse. 14 Wire Gauges are the most common size for lighting. You could get them at your local Home Depot or Amazon if you don't wanna drive.

Now that we have gone over the tools and materials, we have to go over the basics of how our electrical current works. Let's go over what a single-pole switch is, what wires are connected to your switch, and how the electricity flows through the circuit. A single-pole switch is the most basic electrical switch with one input and one output. It's the easiest switch for beginners to use, and that'll be the switch I'll be teaching you how



to wire. A single pole switch typically has one green screw for the green wire, one screw for the input which is where the power goes to, and one screw for the output which connects to the light. The three main wires you'll be using are the green, white, and black wires. The green wires are the Ground wires whose only purpose is to make sure that electrical currents flow to the earth and don't kill you. The white wires are the neutral wires, and they complete the electrical circuit by providing a return path from the light to the power source. And the black wires are the hot wires as they carry the electricity from the breaker to the switch or light source. Electricity flows like a loop. It flows through the

hot wire from the power source to the input screw, then it'll flow out from the output screw to the lightbulb, and then, lastly the lightbulb will return the power through the neutral wire.

Now that we're done with all the technical stuff, we can start the fun. The first thing you need to do is turn off your power supply by accessing your breaker. If you don't turn it off, you'll get shocked, or you might get electrocuted from the ongoing power, and you'll be dead. Next, connect your wires to the power supply. White should go with white, black should go with black, and green should go with green. That's pretty easy. You need

to make sure some copper is showing from your wire. If any wires are fully covered, use your wire stripper to expose at least half an inch of copper at the end of each wire. Every time you connect wires, you'll need to secure them tightly with a wire cap. Unless you're blind, this part should be simple if you're following along. The neutral wire, aka the white wire, should connect directly to the light. Now connect the green wire to the green screw on the switch. Take your wire stripper and strip off enough cover to loop your copper in a hook. Now attach your wire to the green screw and tighten the green screw as tight as you could. Do the same process but now connect the black wire to the input screw on the switch. Now use a new black wire to hook the switch's output screw and connect it to the lightbulb. Lastly, but not least, turn on the power and test the switch to ensure the light bulb turns on.

Now you should have a basic understanding of how to wire a single-pole switch to the light bulb. Today we discussed the materials needed, the fundamentals of how electricity flows through electric currents, and how to wire your single pole switch to your light! Next time you're facing faulty wiring, you could save the day yourself with the right tools and knowledge. And especially you don't have to pay 80-150 bucks and wait for an electrician to work an hour on wiring a switch when you could do it yourself in less than half an hour!

Why Do We Dream?

Darren Wang

Have you ever had a dream or dreamed about anything before? Like sometimes you're riding a dinosaur and then you're late for school with no pants on. Dreams can be funny or scary and also random, and most of the time, they don't make much sense. But there's actually some science behind it. Experts think we dream because it helps our brain sort out memories, deal with our feelings, and respond to brain activity while we are asleep.

One big reason why we dream is to help our brains remember things. Every day, we learn a lot, especially at school. Our brain needs time to go over all that stuff and decide what's important and valuable to remember. Some scientists say dreams help with that. For example, if you studied for a test or played basketball after school, your dream might include something about those things. It's kind of like your brain is doing a clean-up and dreams are part of the process.

Another reason could be to deal with emotions or feelings. If you've ever felt nervous or upset during the day, those feelings can show up in your dreams. I once dreamed I lost my homework right before the due date and we had to go over the homework answers in class—and I was super stressed about school that entire week. That's not a coincidence. Our brains might be helping us work through stuff without us even knowing it. It's all part of the dreaming process.

A third reason is that our brains are just really active while we sleep. During a stage of sleep called REM also known as Rapid Eye Movement, our brain is firing off signals like crazy. Some people believe that dreams are just our brain trying to turn those random moments into a story. That's probably why dreams can be so random, like being chased by a talking banana or swimming in the air.

To wrap it up, even though dreams can be confusing or totally weird, they actually help us in a few important ways. Such as helping us keep memories, understand our feelings, and deal with all the activity happening in our brains while we sleep. So next time you wake up from a wild dream or a scary moment, just remember it might have helped your brain more than you think in many different ways.



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Photoelectric Effect

Kenny Wu

The photoelectric effect is a prominent concept in the field of physics, it occurs when light shines on a metal surface and causes the ejection of electrons. It was first discovered in the late 19th century, and greatly bewildered scientists at the time. This is because the classical wave theory of light could not explain why light below a certain frequency would fail to eject electrons, regardless of its intensity. In 1906, Albert Einstein sparked a breakthrough, proposing that light is made of discrete energy packets, also known as photons, laying the foundation for quantum theory.

Einstein's explanation suggested that energy of each photon is proportional to its frequency, rather than its intensity. The photon can knock an electron loose with an abundant amount of energy. This theory explained why increasing the intensity of a low frequency light would fail to produce any photoelectrons, while a high frequency light could do so despite being dim. Einstein's contribution earned him the Nobel Prize in Physics in 1921.

The emergence of photoelectric effect resulted in extensive implications. It supported the idea that light behaves not only as a wave, but also as a particle. Additionally, it challenged classical physics and introduced the dual nature of light, which would later be validated through other quantum experiments. The introduction of the effect revolutionized our understanding of energy, matter, and the behavior of particles at the atomic scale.

Today, the photoelectric effect is practically utilized in areas such as solar panels, photo detectors, and even automatic doors. The concept remains a cornerstone topic in physics, bridging classical and quantum worlds.

Understanding DNA Crossover During Meiosis

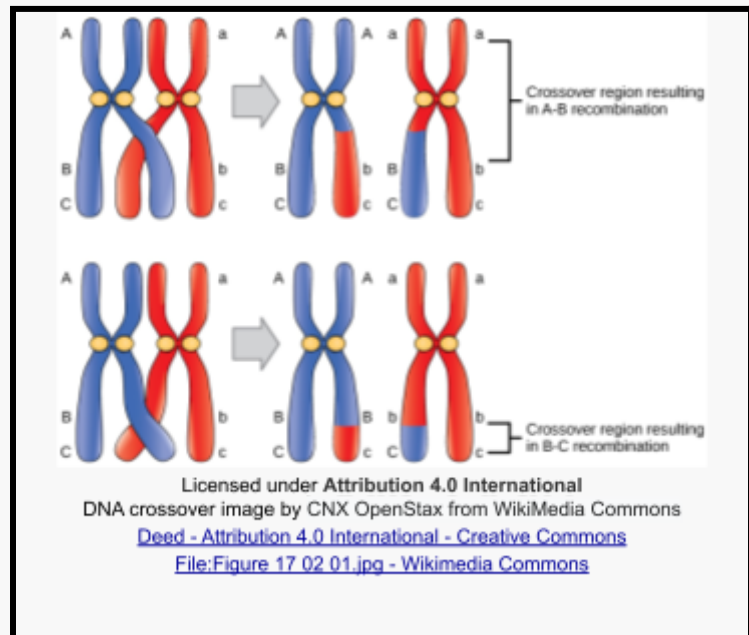
Owen Chen

Crossover is a crucial event that occurs during meiosis, the process by which sex cells—such as sperm and egg cells—are formed. It involves the exchange of genetic material between homologous chromosomes, leading to new combinations of genes that increase genetic diversity in offspring. This mechanism is one of the reasons why siblings can have different traits even though they share the same parents.

During meiosis, a cell goes through two rounds of division to produce four daughter cells, each with half the number of chromosomes as the original. Crossover happens during the first division, specifically in a stage

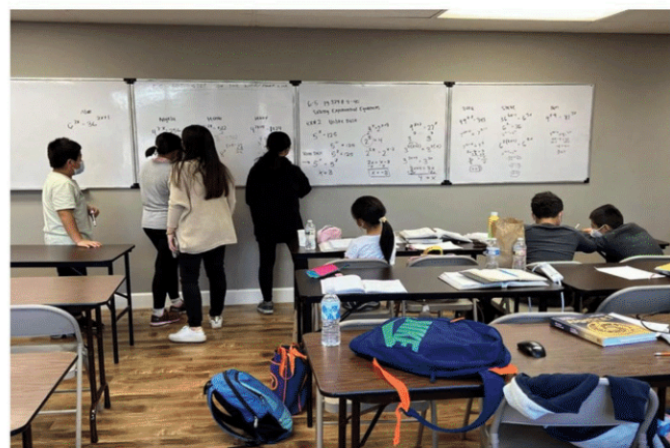
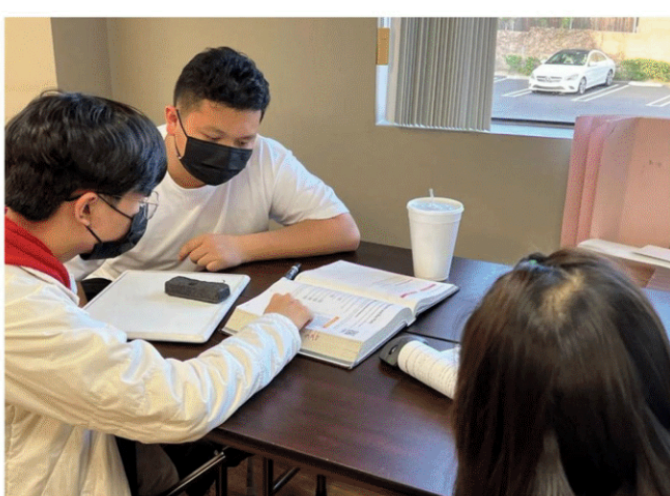
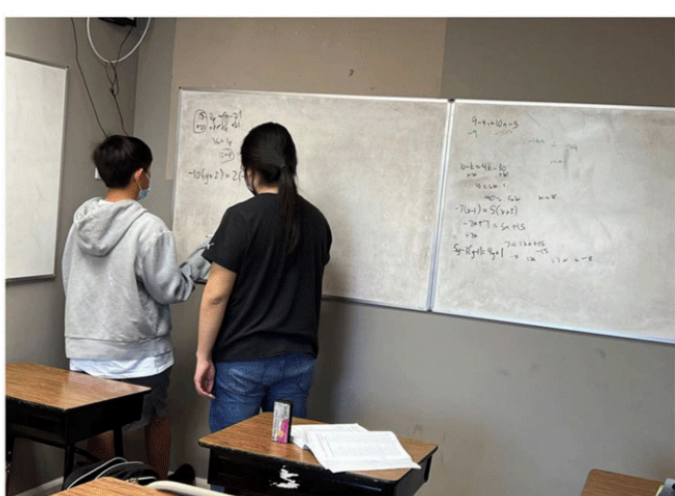
called prophase I. At this point, each chromosome pairs up with its matching chromosome from the other parent. These matched chromosomes are called homologous chromosomes because they carry the same types of genes in the same order, although the versions of those genes—called alleles—might differ.

Once paired, the homologous chromosomes align very closely, forming structures called tetrads. Enzymes then facilitate the physical breakage and rejoining of DNA strands between these chromosomes at specific points called chiasmata. This results in



segments of DNA being swapped between the chromatids, which are the individual arms of the chromosomes. The process is highly controlled to ensure that the exchanges are equal and that the chromosome structure remains intact.

The outcome of crossover is that each resulting sex cell carries a unique set of genetic information. This genetic variation is vital for evolution, as it provides populations with the ability to adapt to changing environments over time. Scientists continue to study crossover not only to understand inheritance patterns but also to uncover causes of genetic disorders that can occur if the process goes wrong. Through crossover, nature ensures that no two individuals are genetically identical, except for identical twins.



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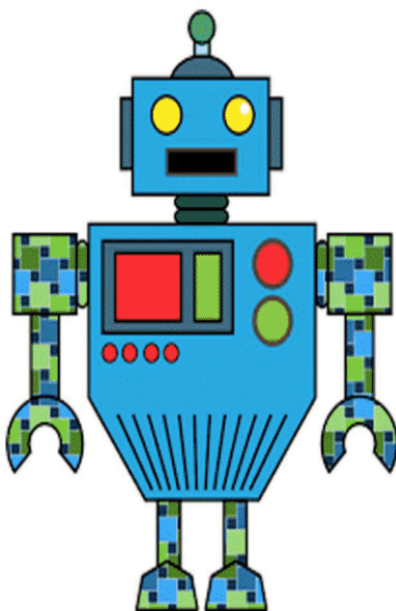
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